

An isometric iteration without a weak limit

Counterexample to TLMC 00000008841

ziangni-sys

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Abstract

The reflection $T(x) = -x$ on the real Hilbert space is asymptotically nonexpansive with all coefficients equal to one. It has a fixed point and preserves a nonempty closed bounded convex interval, yet its Picard orbit from 1 does not converge weakly. This disproves the claimed universal weak convergence of such iterations.

1 Claim and definitions

The source describes “Iteration of asymptotically nonexpansive maps” and asserts that weak convergence of asymptotically nonexpansive sequences always holds in Hilbert space. Its further clause concerns strong convergence and the demiclosedness principle. We disprove the universal weak-convergence assertion, which suffices to disprove the conjunction.

For a map $T : C \rightarrow C$ in a normed space, asymptotic nonexpansiveness means that there are real numbers $k_n \geq 1$, with $k_n \rightarrow 1$, such that

$$\|T^n x - T^n y\| \leq k_n \|x - y\| \quad (x, y \in C, n \geq 0).$$

Including $n = 0$ does not change the standard definition. The iteration is the Picard iteration $x_{n+1} = T x_n$, or $x_n = T^n x_0$. Weak convergence $x_n \rightharpoonup a$ means that $f(x_n) \rightarrow f(a)$ for every continuous linear functional f on the ambient space. No averaging scheme or asymptotic regularity hypothesis is stated in the source.

2 The counterexample

Let $H = \mathbb{R}$ with its usual real Hilbert space structure, let $C = [-1, 1]$, and define $T(x) = -x$ on all of H . Its restriction maps C to itself. The interval is nonempty, closed, bounded, and convex; also $0 \in C$ is a fixed point of T and $x_0 = 1 \in C$.

Proposition 1. *Every iterate of T is an isometry. The orbit $x_n = T^n 1$ lies in C and has no weak limit in H .*

Proof. Negation preserves distance, and a composition of isometries is an isometry. Induction therefore gives

$$|T^n x - T^n y| = |x - y| \quad (n \geq 0).$$

Consequently T is asymptotically nonexpansive with $k_n = 1$ for every n . These same coefficients work for the restriction to C .

Since $T(Tx) = x$, induction on m gives the actual orbit identities

$$x_{2m} = 1, \quad x_{2m+1} = -1 \quad (m \geq 0).$$

In particular the orbit remains in C . If it converged weakly to some $a \in \mathbb{R}$, applying the continuous linear identity functional $f(x) = x$ would give $x_n \rightarrow a$ in $\mathbb{R}$. The even and odd index maps both tend to infinity, so both subsequences would have limit a . They are constant with respective limits 1 and -1 . Uniqueness of real limits would imply $a = 1 = -1$, a contradiction. $\square$

This failure occurs despite boundedness, convexity, closedness, and existence of a fixed point. In particular the demiclosedness clause cannot restore the already false universal weak-convergence assertion. The example does not purport to refute results that assume asymptotic regularity or use an averaged iteration: here $|x_{n+1} - x_n| = 2$ for every n .

3 Lean verification and correspondence

The accompanying Lean 4.19.0 project uses Mathlib at commit `c44e0c8ee63ca166450922a373c7409c5d26b00b`. In namespace `AsymptoticIteration8841`, `reflection` is real negation and `orbit` is defined by Lean’s actual function iteration, not by a supplied table of values. `AsymptoticallyNonexpansive` is the quantified norm inequality above, expressed with metric distance, and `WeaklyConverges` quantifies over all actual continuous real linear functionals. The use of this definition on $\mathbb{R}$ is exactly the standard definition of weak convergence.

`iterate_isometry` proves the isometry assertion for every iterate.

`reflection_asymptotically_nonexpansive` supplies the constant coefficients and their convergence. `interval_geometry` checks the closed, bounded, convex interval and membership of 0 and 1. `orbit_even` and `orbit_odd` prove the orbit identities; the cofinality lemmas and uniqueness of limits then prove `orbit_not_weakly_convergent`. The final theorem `conjecture_00000008841` packages the counterexample, its invariant interval and fixed point, and the nonexistence of a weak limit.

The proof uses no numerical approximation, unproved axiom, or auxiliary computation. Its axiom audit reports only the standard logical axioms `propext`, `Classical.choice`, and `Quot.sound`. Build with `lake build`; check the source and axiom audit with `lake env lean Main.lean` from the `lean` directory.